



WEEK 3 REPORT

SOC INTERNSHIP

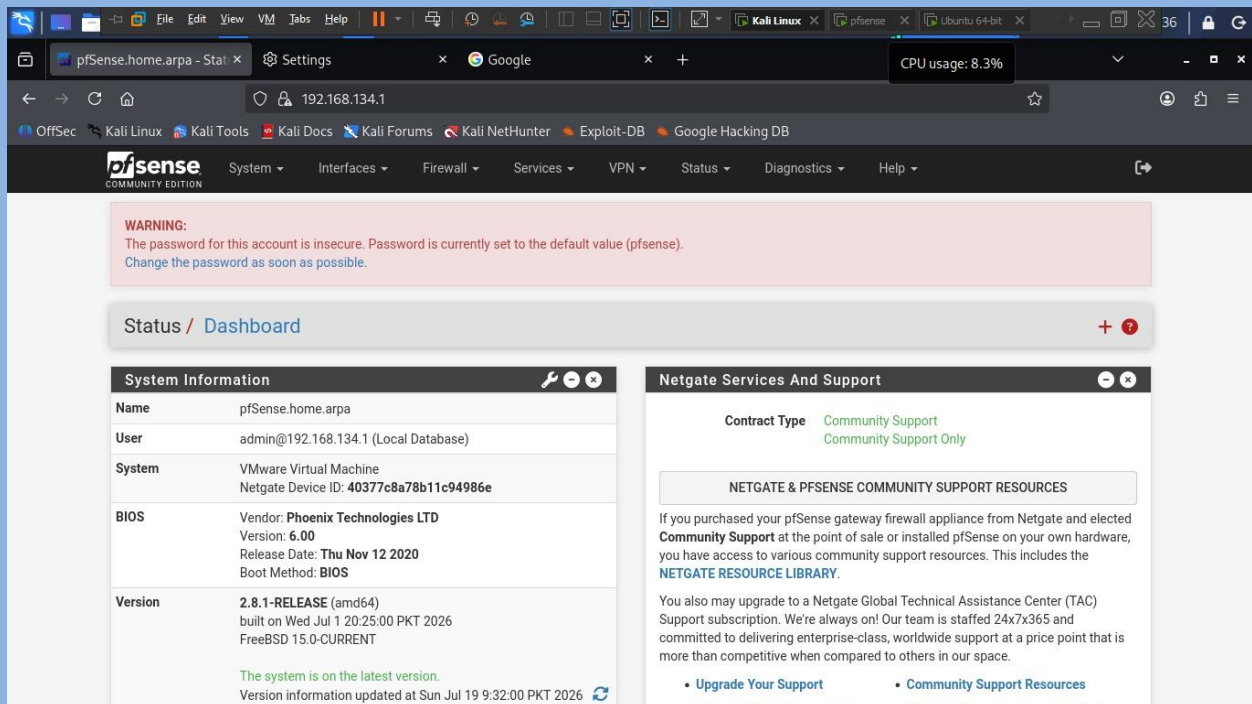
Submitted By: Furqan Ahmed
Registration Number: SOCB01-2273



Task 1: Deploy and Configure pfSense Firewall

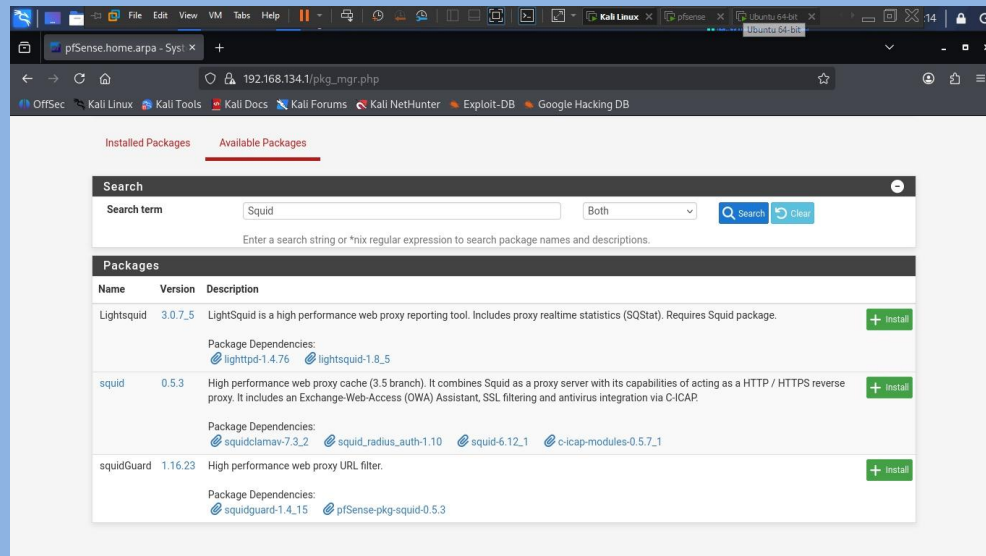
In this task, I deployed **pfSense Firewall** in VMware to secure and manage the network traffic between different virtual machines. First, I assigned the WAN and LAN interfaces. The WAN interface was connected to the NAT network to provide internet access, while the LAN interface was connected to the Host-Only network to allow communication between pfSense, Kali Linux, and the Ubuntu Wazuh server.

After completing the initial setup, I accessed the pfSense web interface using the LAN IP address through a web browser. From the dashboard, I verified that both interfaces were active and working correctly. This firewall acts as the central device that monitors, filters, and controls all network traffic in the lab environment.

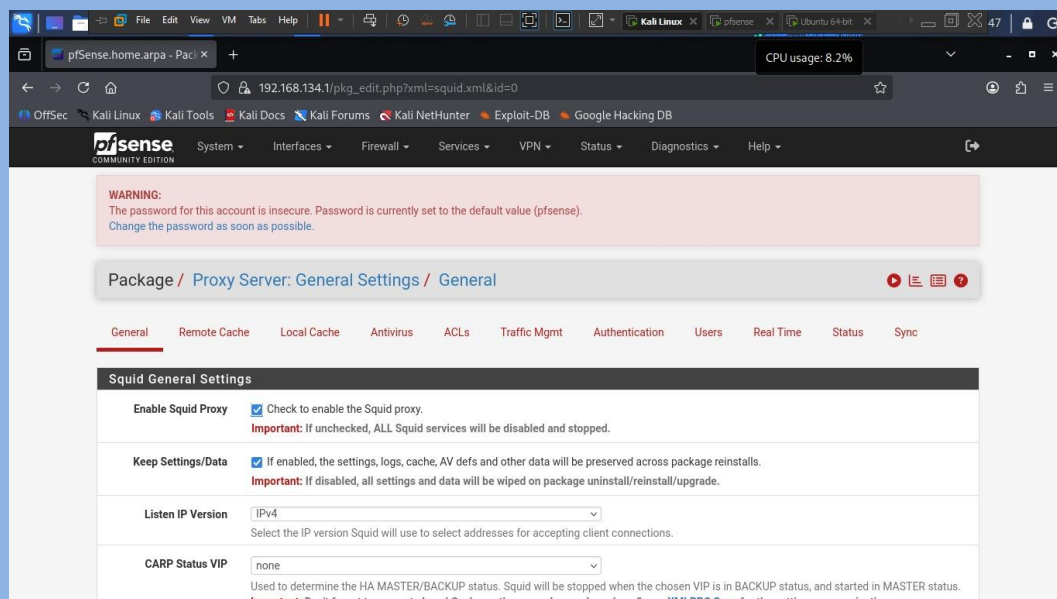


Task 2: Configure Squid Proxy Server

After configuring pfSense, I installed and enabled the **Squid Proxy Server** from the Package Manager. Squid was configured to run on the LAN interface so that client traffic could pass through the proxy server.

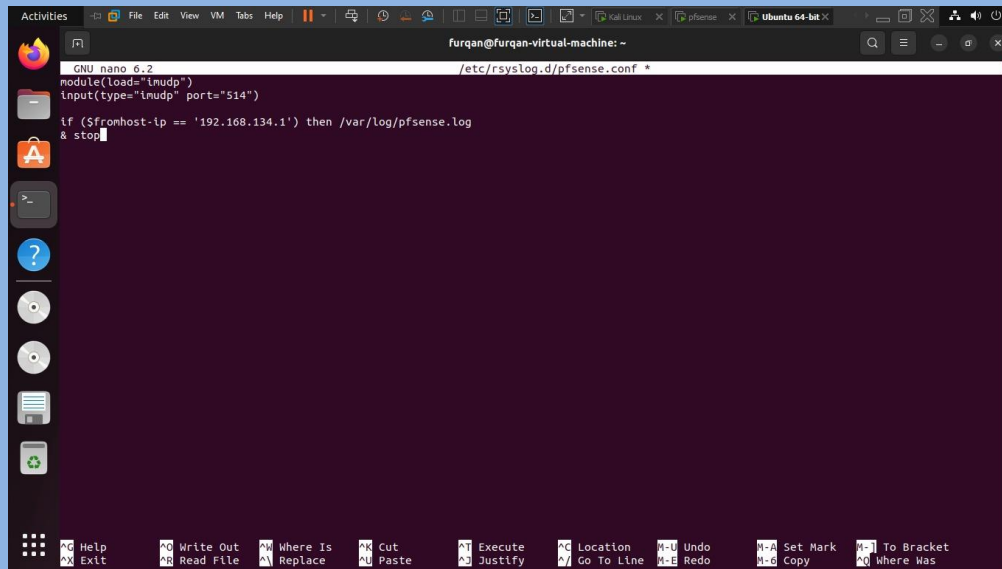


The proxy service was started successfully, and its settings were verified from the Squid General Configuration page. This configuration allows web traffic to be monitored and managed through the proxy server.



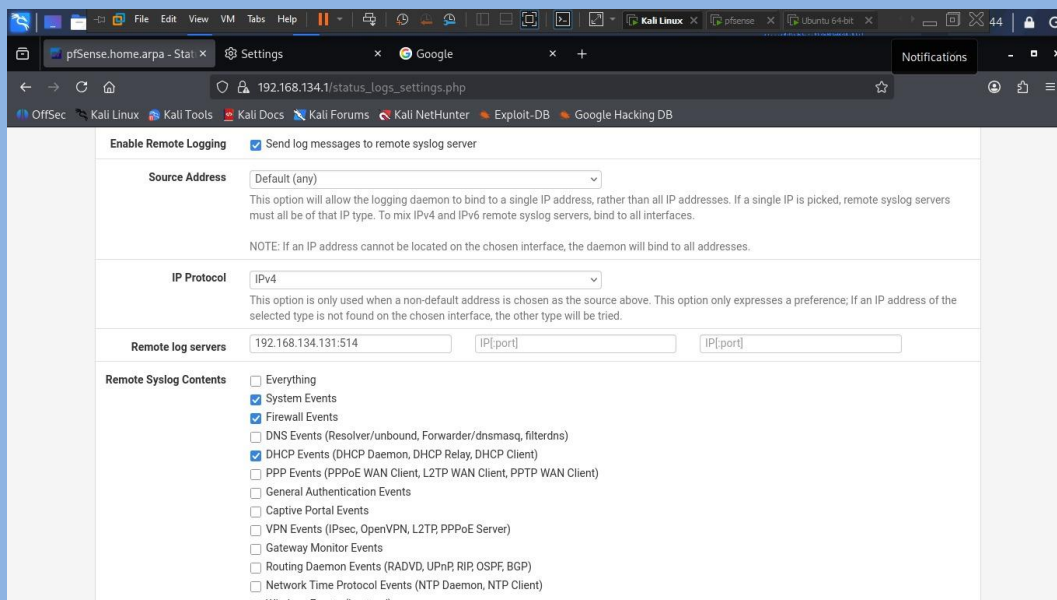
Task 3: Configure Remote Syslog

To centralize firewall logs, I configured **Remote Syslog** in pfSense. The IP address of the Ubuntu Wazuh server (**192.168.134.131**) was entered as the remote log server using UDP port **514**.



```
furqan@furqan-virtual-machine: ~  
GNU nano 6.2 /etc/rsyslog.d/pfsense.conf *  
module(load="imudp")  
input(type="imudp" port="514")  
  
if ($fromhost-ip == '192.168.134.1') then /var/log/pfsense.log  
& stop
```

Firewall Events, DHCP Events, and System Events were selected so that these logs could be forwarded automatically to the Wazuh Manager for security monitoring.



pfSense.home.arpa - Settings

192.168.134.1/status_logs_settings.php

Enable Remote Logging ☒ Send log messages to remote syslog server

Source Address: Default (any)

IP Protocol: IPv4

Remote log servers: 192.168.134.131:514

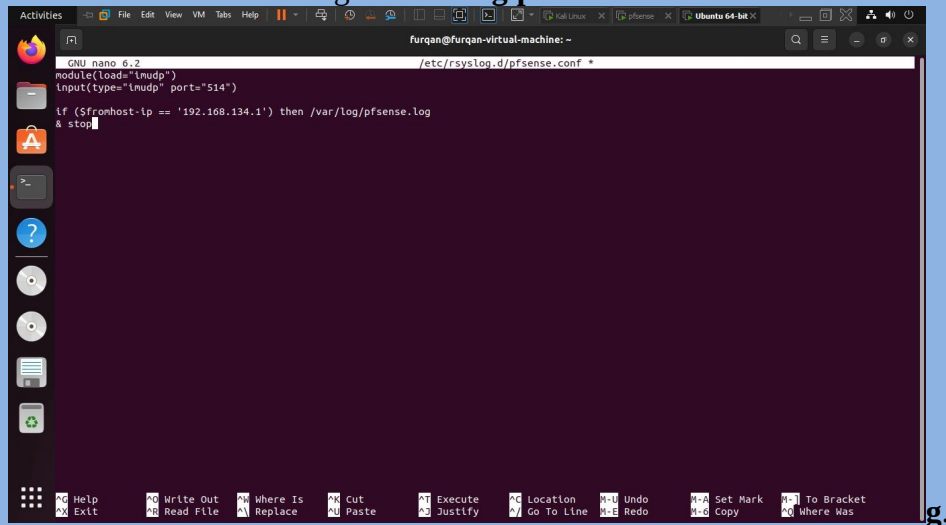
Remote Syslog Contents:

- ☐ Everything
- ☒ System Events
- ☒ Firewall Events
- ☐ DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns)
- ☒ DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client)
- ☐ PPP Events (PPPoE WAN Client, L2TP WAN Client, PPTP WAN Client)
- ☐ General Authentication Events
- ☐ Captive Portal Events
- ☐ VPN Events (IPsec, OpenVPN, L2TP, PPPoE Server)
- ☐ Gateway Monitor Events
- ☐ Routing Daemon Events (RADVD, UPhP, RIP, OSPF, BGP)
- ☐ Network Time Protocol Events (NTP Daemon, NTP Client)
- ☐ Wireless Events (hostapd)

This configuration allows all important events generated by pfSense to be collected on the SIEM server.

Task 4: Configure rsyslog on Ubuntu

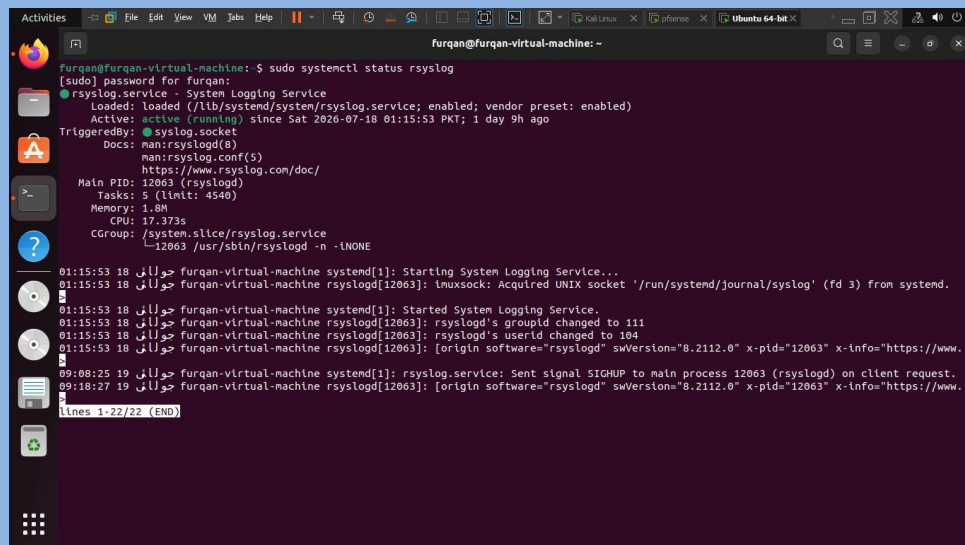
On the Ubuntu machine, I configured **rsyslog** to receive Syslog messages from pfSense. UDP port **514** was enabled, and a separate configuration file was created to store incoming pfSense logs in **/var/log/pfsense.log**



```
GNU nano 6.2 /etc/rsyslog.d/pfsense.conf
module(load="imudp")
input(type="imudp" port="514")

if ($fromhost-ip == '192.168.134.1') then /var/log/pfsense.log
& stop
```

After restarting the rsyslog service, I verified that the service was running successfully and receiving remote logs.



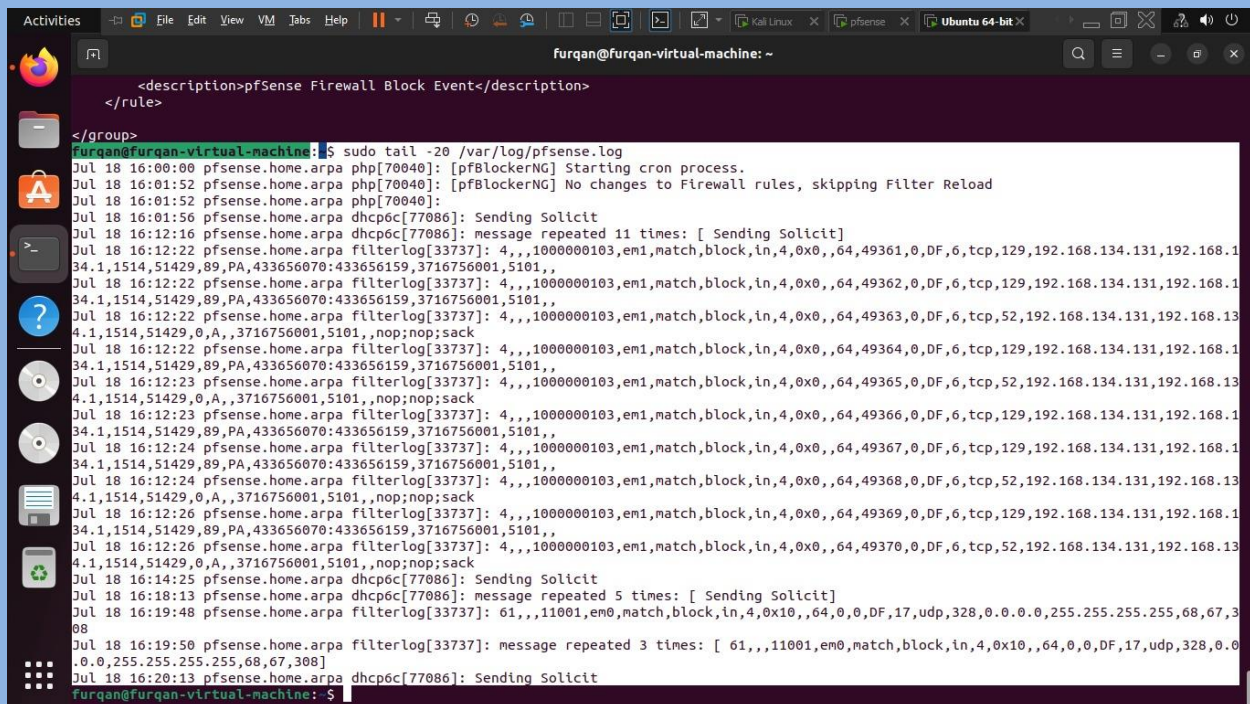
```
furqan@furqan-virtual-machine:~$ sudo systemctl status rsyslog
[sudo] password for furqan:
● rsyslog.service - System Logging Service
   Loaded: loaded (/lib/systemd/system/rsyslog.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-07-18 01:15:53 PKT; 1 day 9h ago
     TriggeredBy: ● syslog.socket
       Docs: man:rsyslogd(8)
            man:rsyslog.conf(5)
            https://www.rsyslog.com/doc/
   Main PID: 12063 (rsyslogd)
      Tasks: 5 (limit: 4540)
     Memory: 1.8M
        CPU: 17.373s
    CGroup: /system.slice/rsyslog.service
            └─12063 /usr/sbin/rsyslogd -n -lNONE

01:15:53 18 جولائی furqan-virtual-machine systemd[1]: Starting System Logging Service...
01:15:53 18 جولائی furqan-virtual-machine rsyslogd[12063]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd.
01:15:53 18 جولائی furqan-virtual-machine systemd[1]: Started System Logging Service.
01:15:53 18 جولائی furqan-virtual-machine rsyslogd[12063]: rsyslogd's groupid changed to 111
01:15:53 18 جولائی furqan-virtual-machine rsyslogd[12063]: rsyslogd's userid changed to 104
01:15:53 18 جولائی furqan-virtual-machine rsyslogd[12063]: [origin software="rsyslogd" swVersion="8.2112.0" x-pid="12063" x-info="https://www.
09:08:25 19 جولائی furqan-virtual-machine systemd[1]: rsyslog.service: Sent signal SIGCHUP to main process 12063 (rsyslogd) on client request.
09:18:27 19 جولائی furqan-virtual-machine rsyslogd[12063]: [origin software="rsyslogd" swVersion="8.2112.0" x-pid="12063" x-info="https://www.
lines 1-22/22 (END)
```


Task 5: Verify pfSense Logs

To confirm that the Syslog configuration was working correctly, I monitored the **pfsense.log** file using the `tail` command. The log file successfully displayed DHCP events and firewall block events received from pfSense.

This confirmed that log forwarding between pfSense and the Ubuntu server was functioning correctly.

A screenshot of a terminal window on a virtual machine named 'furqan@furqan-virtual-machine: ~'. The terminal shows the command `sudo tail -20 /var/log/pfsense.log` being executed. The output displays various log entries from pfSense, including DHCP events and firewall block events. The logs are timestamped with dates like 'Jul 18' and times like '16:00:00'. The entries include details about IP addresses, ports, and the actions taken by the firewall. The terminal window has a dark background with a light-colored text area. The top of the window shows the Ubuntu desktop environment with various icons and a menu bar.

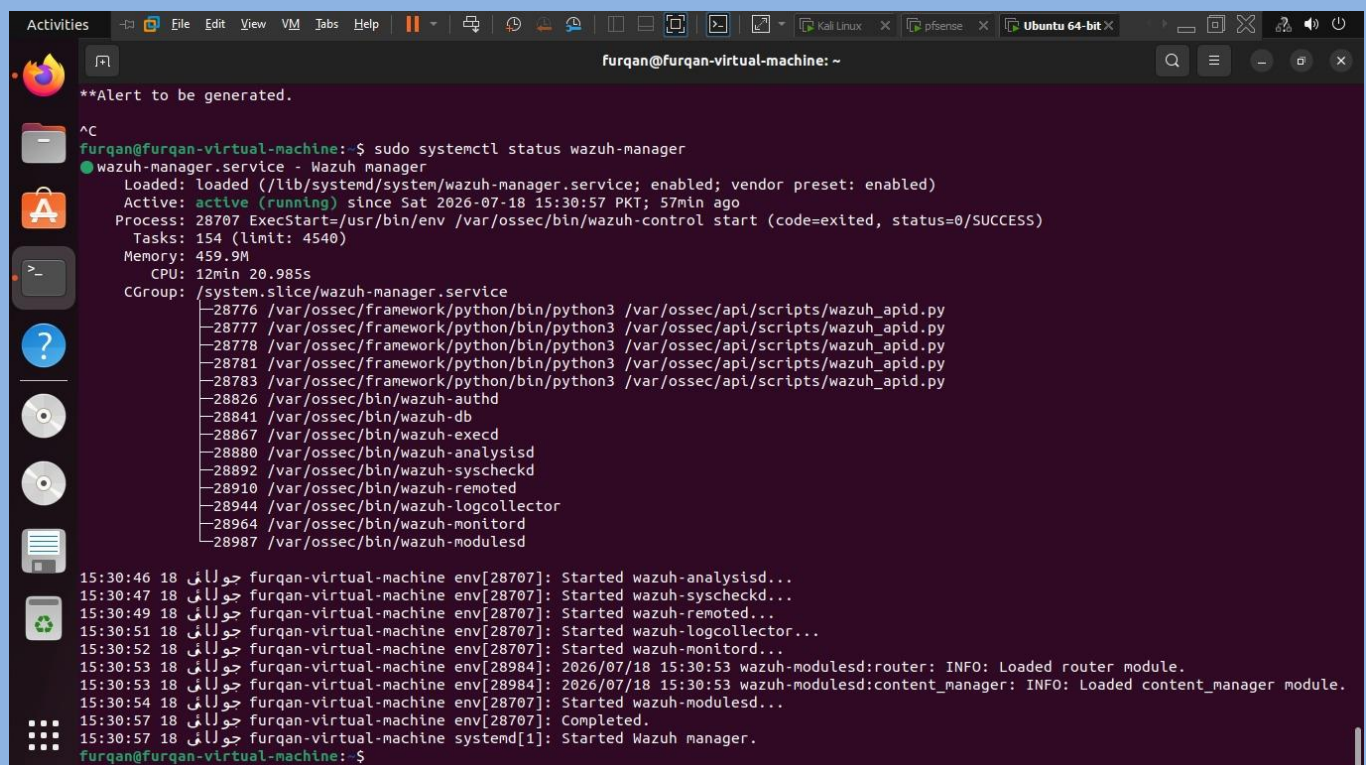
```
<description>pfsense Firewall Block Event</description>
</rule>
</group>
furqan@furqan-virtual-machine:~$ sudo tail -20 /var/log/pfsense.log
Jul 18 16:00:00 pfsense.home.arpa php[70040]: [pfBlockerNG] Starting cron process.
Jul 18 16:01:52 pfsense.home.arpa php[70040]: [pfBlockerNG] No changes to Firewall rules, skipping Filter Reload
Jul 18 16:01:52 pfsense.home.arpa php[70040]:
Jul 18 16:01:56 pfsense.home.arpa dhcp6c[77086]: Sending Solicit
Jul 18 16:12:16 pfsense.home.arpa dhcp6c[77086]: message repeated 11 times: [ Sending Solicit]
Jul 18 16:12:22 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49361,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,51429,89,PA,433656070:433656159,3716756001,5101,,
Jul 18 16:12:22 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49362,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,51429,89,PA,433656070:433656159,3716756001,5101,,
Jul 18 16:12:22 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49363,0,DF,6,tcp,52,192.168.134.131,192.168.134.1,1514,51429,0,A,,3716756001,5101,,nop;nop;sack
Jul 18 16:12:22 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49364,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,51429,89,PA,433656070:433656159,3716756001,5101,,
Jul 18 16:12:23 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49365,0,DF,6,tcp,52,192.168.134.131,192.168.134.1,1514,51429,0,A,,3716756001,5101,,nop;nop;sack
Jul 18 16:12:23 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49366,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,51429,89,PA,433656070:433656159,3716756001,5101,,
Jul 18 16:12:24 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49367,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,51429,89,PA,433656070:433656159,3716756001,5101,,
Jul 18 16:12:24 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49368,0,DF,6,tcp,52,192.168.134.131,192.168.134.1,1514,51429,0,A,,3716756001,5101,,nop;nop;sack
Jul 18 16:12:26 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49369,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,51429,89,PA,433656070:433656159,3716756001,5101,,
Jul 18 16:12:26 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,49370,0,DF,6,tcp,52,192.168.134.131,192.168.134.1,1514,51429,0,A,,3716756001,5101,,nop;nop;sack
Jul 18 16:14:25 pfsense.home.arpa dhcp6c[77086]: Sending Solicit
Jul 18 16:18:13 pfsense.home.arpa dhcp6c[77086]: message repeated 5 times: [ Sending Solicit]
Jul 18 16:19:48 pfsense.home.arpa filterlog[33737]: 61,,,11001,em0,match,block,in,4,0x10,,64,0,0,DF,17,udp,328,0.0.0.0,255.255.255.255,68,67,308
Jul 18 16:19:50 pfsense.home.arpa filterlog[33737]: message repeated 3 times: [ 61,,,11001,em0,match,block,in,4,0x10,,64,0,0,DF,17,udp,328,0.0.0.0,255.255.255.255,68,67,308]
Jul 18 16:20:13 pfsense.home.arpa dhcp6c[77086]: Sending Solicit
furqan@furqan-virtual-machine:~$
```

Task 6: Configure Wazuh Log Collection

Theory

The Wazuh Manager was configured to monitor the **/var/log/pfsense.log** file by adding it to the **ossec.conf** configuration file. This allowed the Wazuh Log Collector to continuously read firewall logs received from pfSense..

```
furqan@furqan-virtual-machine:~$ sudo grep -A3 -B2 pfsense /var/ossec/etc/ossec.conf
<localfile>
<location>/var/log/pfsense.log</location>
<log_format>syslog</log_format>
</localfile>
furqan@furqan-virtual-machine:~$
```

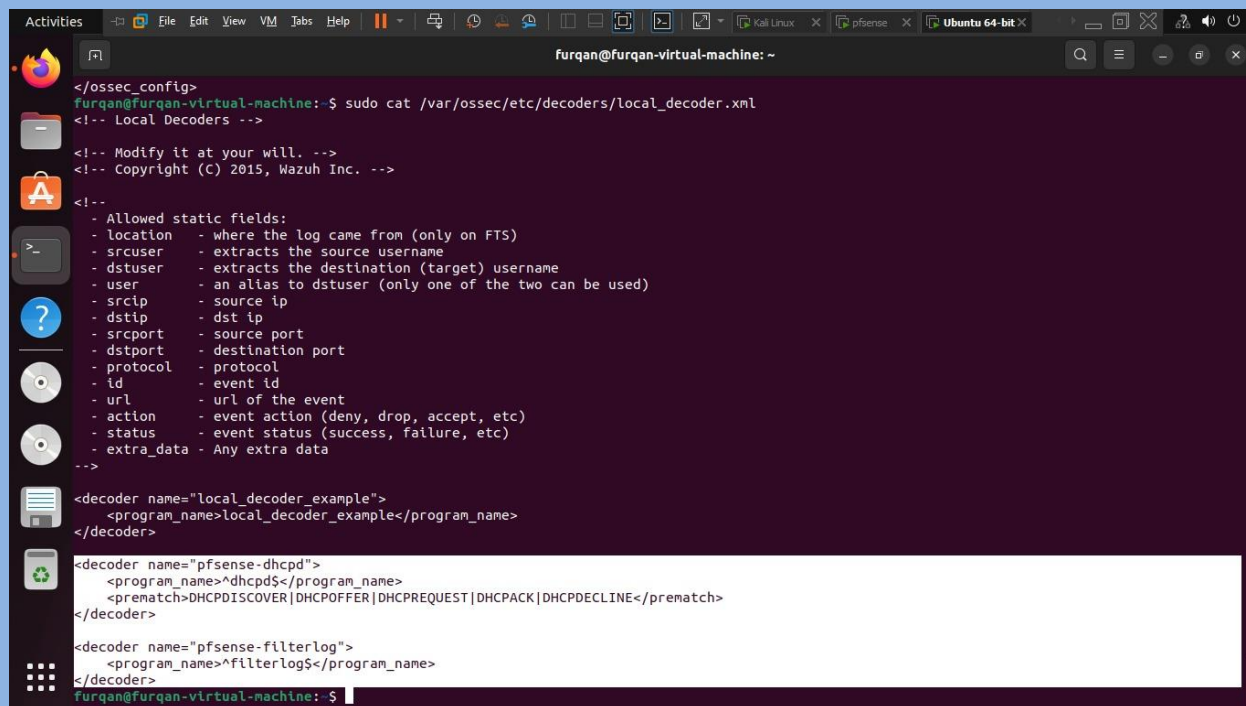


```
Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
**Alert to be generated.
^C
furqan@furqan-virtual-machine:~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-07-18 15:30:57 PKT; 57min ago
     Process: 28707 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 154 (limit: 4540)
   Memory: 459.9M
      CPU: 12min 20.985s
   CGroup: /system.slice/wazuh-manager.service
           └─28776 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─28777 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─28778 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─28781 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─28783 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─28826 /var/ossec/bin/wazuh-authd
                       └─28841 /var/ossec/bin/wazuh-db
                         └─28867 /var/ossec/bin/wazuh-execd
                           └─28880 /var/ossec/bin/wazuh-analysisd
                             └─28892 /var/ossec/bin/wazuh-syscheckd
                               └─28910 /var/ossec/bin/wazuh-remoted
                                 └─28944 /var/ossec/bin/wazuh-logcollector
                                   └─28964 /var/ossec/bin/wazuh-monitord
                                     └─28987 /var/ossec/bin/wazuh-modulesd

15:30:46 18 جولائی furqan-virtual-machine env[28707]: Started wazuh-analysisd...
15:30:47 18 جولائی furqan-virtual-machine env[28707]: Started wazuh-syscheckd...
15:30:49 18 جولائی furqan-virtual-machine env[28707]: Started wazuh-remoted...
15:30:51 18 جولائی furqan-virtual-machine env[28707]: Started wazuh-logcollector...
15:30:52 18 جولائی furqan-virtual-machine env[28707]: Started wazuh-monitord...
15:30:53 18 جولائی furqan-virtual-machine env[28984]: 2026/07/18 15:30:53 wazuh-modulesd:router: INFO: Loaded router module.
15:30:53 18 جولائی furqan-virtual-machine env[28984]: 2026/07/18 15:30:53 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
15:30:54 18 جولائی furqan-virtual-machine env[28707]: Started wazuh-modulesd...
15:30:57 18 جولائی furqan-virtual-machine env[28707]: Completed.
15:30:57 18 جولائی furqan-virtual-machine systemd[1]: Started Wazuh manager.
furqan@furqan-virtual-machine:~$
```

Task 7: Create Custom Decoder

To identify DHCP logs received from pfSense, I created a custom decoder named **pfsense-dhcpd** in the **local_decoder.xml** file.



```
</ossec_config>
furqan@furqan-virtual-machine:~$ sudo cat /var/ossec/etc/decoders/local_decoder.xml
<!-- Local Decoders -->

<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!--
- Allowed static fields:
- location - where the log came from (only on FTS)
- srcuser - extracts the source username
- dstuser - extracts the destination (target) username
- user - an alias to dstuser (only one of the two can be used)
- srcip - source ip
- dstip - dst ip
- srcport - source port
- dstport - destination port
- protocol - protocol
- id - event id
- url - url of the event
- action - event action (deny, drop, accept, etc)
- status - event status (success, failure, etc)
- extra_data - Any extra data
-->

<decoder name="local_decoder_example">
  <program_name>local_decoder_example</program_name>
</decoder>

<decoder name="pfsense-dhcpd">
  <program_name>^dhcpd$</program_name>
  <prematch>DHCPDISCOVER|DHCPOFFER|DHCPREQUEST|DHCPACK|DHCPDECLINE</prematch>
</decoder>

<decoder name="pfsense-filterlog">
  <program_name>^filterlog$</program_name>
</decoder>
furqan@furqan-virtual-machine:~$
```

The decoder was configured to recognize logs generated by the **dhcpd** service. This enables Wazuh to correctly identify DHCP-related events before applying detection rules.

Task 8: Create Custom Rule

After creating the decoder, I created a custom Wazuh rule with **Rule ID 100100** in the **local_rules.xml** file.

The rule was configured to generate an alert whenever a DHCP event matched the **pfsense-dhcpd** decoder.

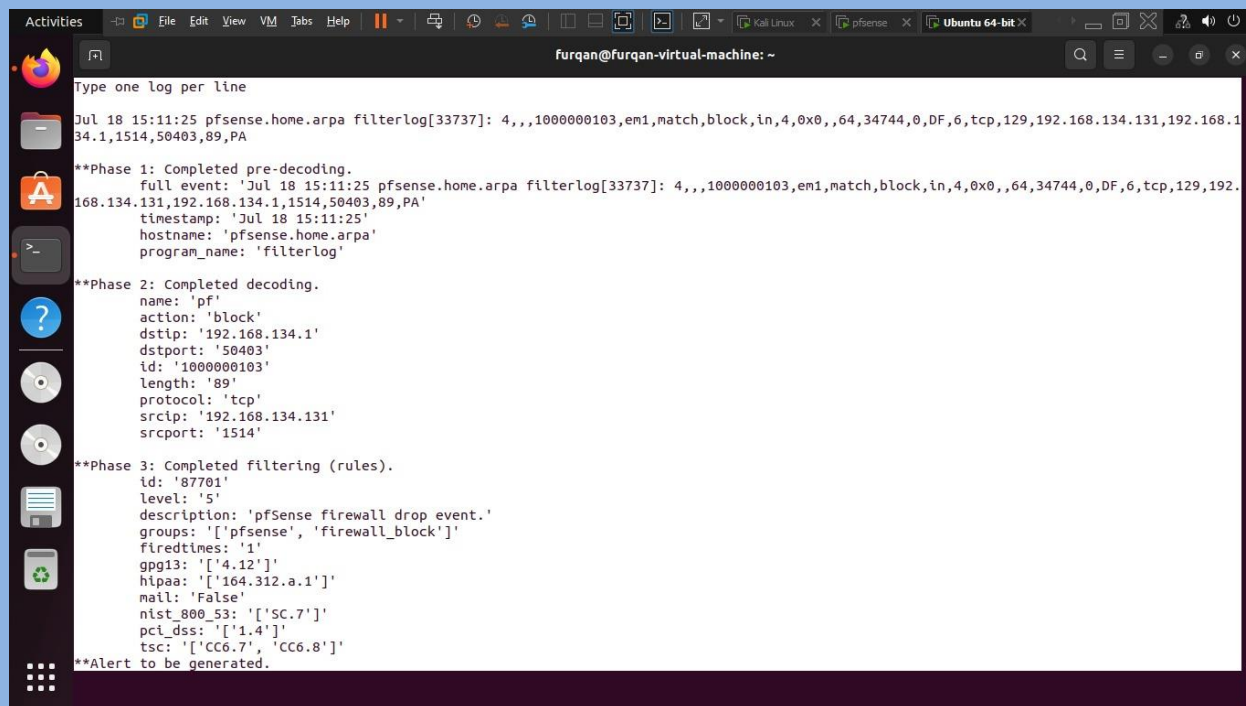

```
Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
<!-- Local rules -->
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->
<!-- Example -->
<group name="local,syslog,sshd,">
  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>
</group>
<group name="pfsense,dhcp,">
  <rule id="100100" level="5">
    <decoded_as>pfsense-dhcpd</decoded_as>
    <description>pfsense DHCP Event Detected</description>
  </rule>
</group>
<group name="pfsense,firewall,">
  <rule id="100101" level="7">
    <decoded_as>pfsense-filterlog</decoded_as>
    <description>pfsense Firewall Block Event</description>
  </rule>
</group>
furqan@furqan-virtual-machine: ~$
```

This customization demonstrates how Wazuh can detect organization-specific events.

```
Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
<!-- Local rules -->
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->
<!-- Example -->
<group name="local,syslog,sshd,">
  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>
</group>
<group name="pfsense,dhcp,">
  <rule id="100100" level="5">
    <decoded_as>pfsense-dhcpd</decoded_as>
    <description>pfsense DHCP Event Detected</description>
  </rule>
</group>
<group name="pfsense,firewall,">
  <rule id="100101" level="7">
    <decoded_as>pfsense-filterlog</decoded_as>
    <description>pfsense Firewall Block Event</description>
  </rule>
</group>
furqan@furqan-virtual-machine: ~$
```

Task 9: Validate Custom Rule

The custom decoder and rule were tested using the **wazuh-logtest** utility. A sample DHCP log generated by pfSense was provided as input.



```
furqan@furqan-virtual-machine: ~  
Type one log per line  
Jul 18 15:11:25 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,34744,0,DF,6,tcp,129,192.168.134.1,192.168.134.1,1514,50403,89,PA  
  
**Phase 1: Completed pre-decoding.  
full event: 'Jul 18 15:11:25 pfsense.home.arpa filterlog[33737]: 4,,,1000000103,em1,match,block,in,4,0x0,,64,34744,0,DF,6,tcp,129,192.168.134.1,192.168.134.1,1514,50403,89,PA'  
timestamp: 'Jul 18 15:11:25'  
hostname: 'pfsense.home.arpa'  
program_name: 'filterlog'  
  
**Phase 2: Completed decoding.  
name: 'pf'  
action: 'block'  
dstip: '192.168.134.1'  
dstport: '50403'  
id: '1000000103'  
length: '89'  
protocol: 'tcp'  
srcip: '192.168.134.131'  
srcport: '1514'  
  
**Phase 3: Completed filtering (rules).  
id: '87701'  
level: '5'  
description: 'pfSense firewall drop event.'  
groups: '['pfsense', 'firewall_block']'  
firedtimes: '1'  
gpg13: '['4.12']'  
hipaa: '['164.312.a.1']'  
mail: 'False'  
nist_800_53: '['SC.7']'  
pci_dss: '['1.4']'  
tsc: '['CC6.7', 'CC6.8']'  
  
**Alert to be generated.
```

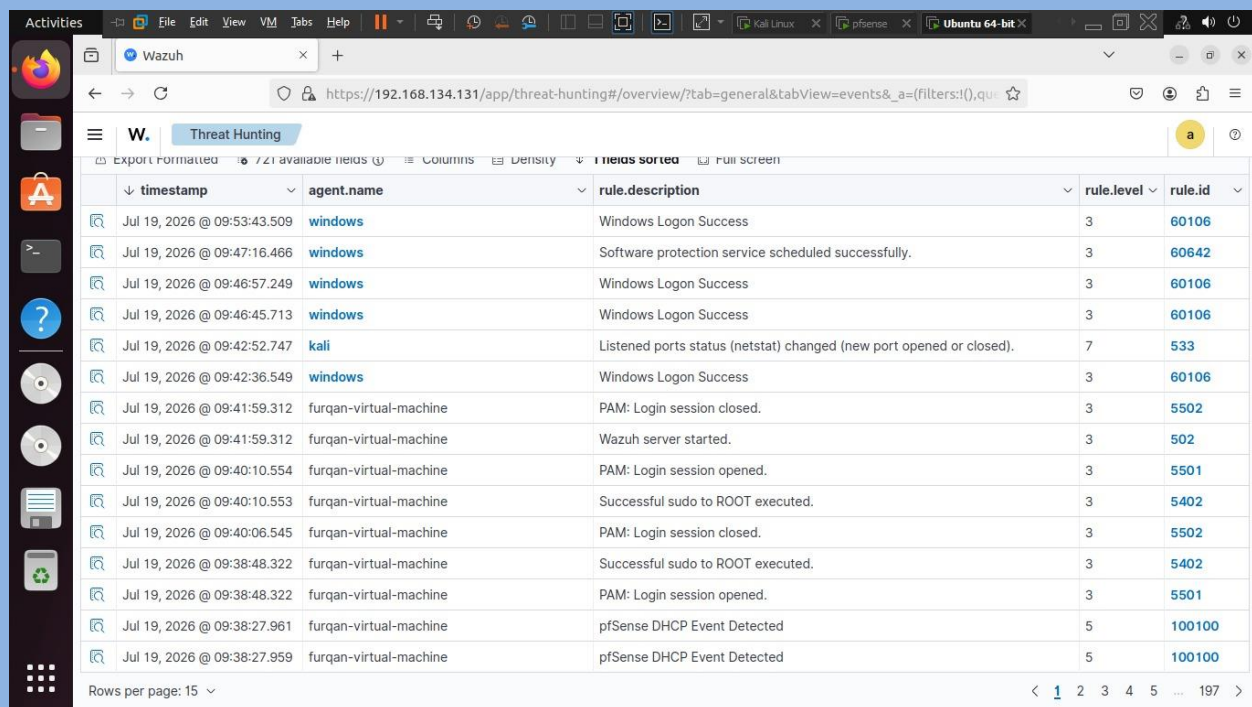
The output successfully completed:

- Phase 1 – Pre-decoding
- Phase 2 – Decoding
- Phase 3 – Rule Matching

Finally, the message "**Alert to be generated**" confirmed that the custom rule was working correctly.

Task 10: Monitor Alerts in Wazuh Dashboard

After successful rule validation, I verified the generated alerts in the **Wazuh Dashboard**. The custom alert with **Rule ID 100100** appeared in the Threat Hunting section, confirming that Wazuh successfully detected the DHCP event received from pfSense.



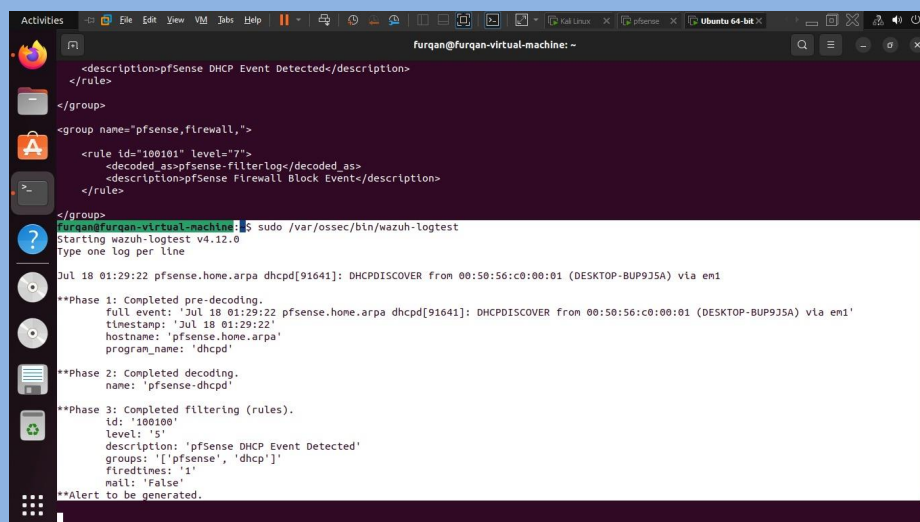
The screenshot shows the Wazuh Threat Hunting interface in a web browser. The URL is `https://192.168.134.131/app/threat-hunting#/overview/?tab=general&tabView=events&_a=(filters:(),query:)`. The interface displays a table of events with columns: timestamp, agent.name, rule.description, rule.level, and rule.id. The table contains 15 rows of data, showing events from various agents including windows, kali, and furqan-virtual-machine. The events include Windows Logon Success, Software protection service scheduled successfully, PAM: Login session closed, Wazuh server started, and pfSense DHCP Event Detected.

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 19, 2026 @ 09:53:43.509	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:47:16.466	windows	Software protection service scheduled successfully.	3	60642
Jul 19, 2026 @ 09:46:57.249	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:46:45.713	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:42:52.747	kali	Listened ports status (netstat) changed (new port opened or closed).	7	533
Jul 19, 2026 @ 09:42:36.549	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:41:59.312	furqan-virtual-machine	PAM: Login session closed.	3	5502
Jul 19, 2026 @ 09:41:59.312	furqan-virtual-machine	Wazuh server started.	3	502
Jul 19, 2026 @ 09:40:10.554	furqan-virtual-machine	PAM: Login session opened.	3	5501
Jul 19, 2026 @ 09:40:10.553	furqan-virtual-machine	Successful sudo to ROOT executed.	3	5402
Jul 19, 2026 @ 09:40:06.545	furqan-virtual-machine	PAM: Login session closed.	3	5502
Jul 19, 2026 @ 09:38:48.322	furqan-virtual-machine	Successful sudo to ROOT executed.	3	5402
Jul 19, 2026 @ 09:38:48.322	furqan-virtual-machine	PAM: Login session opened.	3	5501
Jul 19, 2026 @ 09:38:27.961	furqan-virtual-machine	pfSense DHCP Event Detected	5	100100
Jul 19, 2026 @ 09:38:27.959	furqan-virtual-machine	pfSense DHCP Event Detected	5	100100

This verified the complete workflow from log generation to alert visualization.

Task 11: Firewall Event Detection

Besides the custom DHCP rule, Wazuh also detected built-in pfSense firewall events. Firewall block logs generated by pfSense matched the default Wazuh firewall rule (**Rule ID 87701**), confirming that firewall traffic was being analyzed successfully.



The screenshot shows a terminal window titled "furqan@furqan-virtual-machine: ~". The user has run the command `sudo /var/ossec/bin/wazuh-logtest`. The output shows the Wazuh logtest interface, including the XML configuration for the rule, the log input, and the resulting alert details.

```

<description>pfSense DHCP Event Detected</description>
</rule>
</group>
<group name="pfsense, firewall,">
  <rule id="100100" level="7">
    <decoded_as>pfsense-filterlog</decoded_as>
    <description>pfSense Firewall Block Event</description>
  </rule>
</group>
furqan@furqan-virtual-machine:~$ sudo /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.12.0
Type one log per line

Jul 18 01:29:22 pfsense.hone.arpa dhcpd[91641]: DHCPDISCOVER from 00:50:56:c8:00:01 (DESKTOP-BUP93JA) via eni

**Phase 1: Completed pre-decoding.
full event: 'Jul 18 01:29:22 pfsense.hone.arpa dhcpd[91641]: DHCPDISCOVER from 00:50:56:c8:00:01 (DESKTOP-BUP93JA) via eni'
timestamp: 'Jul 18 01:29:22'
hostname: 'pfsense.hone.arpa'
program_name: 'dhcpd'

**Phase 2: Completed decoding.
name: 'pfsense-dhcpd'

**Phase 3: Completed filtering (rules).
id: '100100'
level: '5'
description: 'pfSense DHCP Event Detected'
groups: '['pfsense', 'dhcpd']'
firedtines: '1'
mail: 'False'

**Alert to be generated.

```

Task 12: Incident Response Plan

Theory

An Incident Response Plan (IRP) is a structured process that helps security teams identify, analyze, contain, and recover from security incidents. It ensures that every incident is handled in a systematic way to minimize its impact and restore normal operations quickly. In this project, the Incident Response Plan was implemented based on the alerts generated by **pfSense** and **Wazuh**.

1. Preparation

In the preparation phase, the required security environment was successfully deployed. pfSense Firewall was configured to monitor network traffic, Squid Proxy Server was installed, and Remote Syslog was enabled to forward firewall logs to the Ubuntu-based Wazuh Manager. Wazuh was configured to collect and analyze logs from `/var/log/pfsense.log`. A custom decoder (`pfsense-dhcpd`) and a custom rule (Rule ID **100100**) were also created to detect DHCP events.

The screenshot displays the pfSense Community Edition web interface. At the top, a navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. A warning message states: "WARNING: The password for this account is insecure. Password is currently set to the default value (pfsense). Change the password as soon as possible." Below this, the "Status / Dashboard" section is visible. It contains two main panels: "System Information" and "Netgate Services And Support".

System Information	
Name	pfSense.home.arpa
User	admin@192.168.134.1 (Local Database)
System	VMware Virtual Machine Netgate Device ID: 40377c8a78b11c94986e
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Thu Nov 12 2020 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Wed Jul 1 20:25:00 PKT 2026 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Sun Jul 19 9:32:00 PKT 2026

Netgate Services And Support

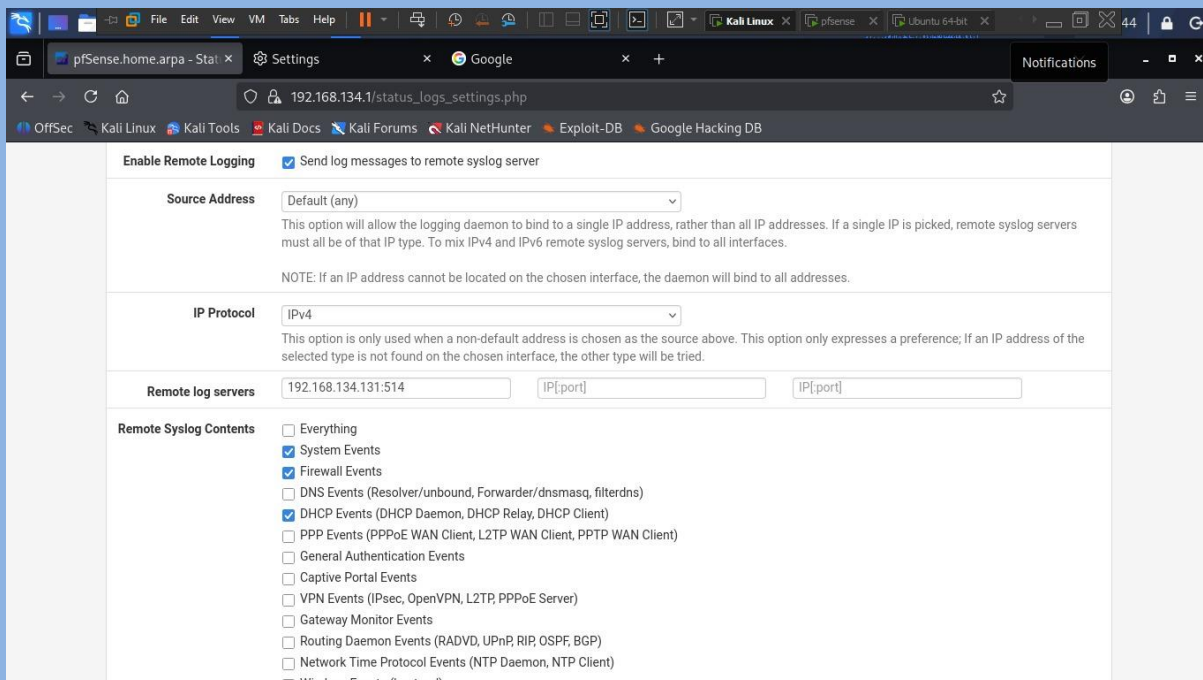
Contract Type: Community Support
Community Support Only

NETGATE & PFSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected **Community Support** at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the **NETGATE RESOURCE LIBRARY**.

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- Upgrade Your Support
- Community Support Resources



```

Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
**Alert to be generated.
^C
furqan@furqan-virtual-machine:~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-07-18 15:30:57 PKT; 57min ago
     Process: 28707 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 154 (limit: 4540)
   Memory: 459.9M
      CPU: 12min 20.985s
   CGroup: /system.slice/wazuh-manager.service
           └─28776 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─28777 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─28778 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─28781 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─28783 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─28826 /var/ossec/bin/wazuh-authd
                       └─28841 /var/ossec/bin/wazuh-db
                         └─28867 /var/ossec/bin/wazuh-execd
                           └─28880 /var/ossec/bin/wazuh-analysisd
                             └─28892 /var/ossec/bin/wazuh-syscheckd
                               └─28910 /var/ossec/bin/wazuh-remoted
                                 └─28944 /var/ossec/bin/wazuh-logcollector
                                   └─28964 /var/ossec/bin/wazuh-monitord
                                     └─28987 /var/ossec/bin/wazuh-modulesd

15:30:46 18 جولي furqan-virtual-machine env[28707]: Started wazuh-analysisd...
15:30:47 18 جولي furqan-virtual-machine env[28707]: Started wazuh-syscheckd...
15:30:49 18 جولي furqan-virtual-machine env[28707]: Started wazuh-remoted...
15:30:51 18 جولي furqan-virtual-machine env[28707]: Started wazuh-logcollector...
15:30:52 18 جولي furqan-virtual-machine env[28707]: Started wazuh-monitord...
15:30:53 18 جولي furqan-virtual-machine env[28984]: 2026/07/18 15:30:53 wazuh-modulesd:router: INFO: Loaded router module.
15:30:53 18 جولي furqan-virtual-machine env[28984]: 2026/07/18 15:30:53 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
15:30:54 18 جولي furqan-virtual-machine env[28707]: Started wazuh-modulesd...
15:30:57 18 جولي furqan-virtual-machine env[28707]: Completed.
15:30:57 18 جولي furqan-virtual-machine systemd[1]: Started Wazuh manager.
furqan@furqan-virtual-machine:~$

```

2. Identification

During the identification phase, firewall logs generated by pfSense were successfully received on the Wazuh server. The logs were monitored using the `tail` command and analyzed by Wazuh. The custom decoder identified DHCP events, while the custom rule generated alerts. In addition, built-in firewall block events were also detected successfully.


```
Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
GNU nano 6.2 /var/ossec/etc/ossec.conf *
<!--
Wazuh - Manager - Default configuration for ubuntu 22.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
<global>
  <jsonout_output>yes</jsonout_output>
  <alerts_log>yes</alerts_log>
  <logall>no</logall>
  <logall_json>no</logall_json>
  <email_notification>no</email_notification>
  <smtp_server>smtp.example.wazuh.com</smtp_server>
  <email_from>wazuh@example.wazuh.com</email_from>
  <email_to>recipient@example.wazuh.com</email_to>
  <email_maxperhour>12</email_maxperhour>
  <email_log_source>alerts.log</email_log_source>
  <agents_disconnection_time>10m</agents_disconnection_time>
  <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
  <update_check>yes</update_check>
</global>

<localfile>
  <location>/var/log/pfsense.log</location>
  <log_format>syslog</log_format>
</localfile>

<alerts>
  <log_alert_level>3</log_alert_level>
  <email_alert_level>12</email_alert_level>
</alerts>

<!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->

Help  Write Out  Where Is  Cut  Execute  Location  M-U Undo  M-A Set Mark  M-T To Bracket
Exit  Read File  Replace  Paste  Justify  Go To Line  M-E Redo  M-C Copy  M-W Where Was
```

```
Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
Type one log per line
Jul 18 15:11:25 pfsense.home.arpa filterlog[33737]: 4,,1000000103,em1,match,block,in,4,0x0,,64,34744,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,50403,89,PA

**Phase 1: Completed pre-decoding.
full event: 'Jul 18 15:11:25 pfsense.home.arpa filterlog[33737]: 4,,1000000103,em1,match,block,in,4,0x0,,64,34744,0,DF,6,tcp,129,192.168.134.131,192.168.134.1,1514,50403,89,PA'
timestamp: 'Jul 18 15:11:25'
hostname: 'pfsense.home.arpa'
program_name: 'filterlog'

**Phase 2: Completed decoding.
name: 'pf'
action: 'block'
dstip: '192.168.134.1'
dstport: '50403'
id: '1000000103'
length: '89'
protocol: 'tcp'
srcip: '192.168.134.131'
srcport: '1514'

**Phase 3: Completed filtering (rules).
id: '87701'
level: '5'
description: 'pfSense firewall drop event.'
groups: '['pfsense', 'firewall_block']'
firedtimes: '1'
gpg13: '['4.12']'
hipaa: '['164.312.a.1']'
mail: 'False'
nist_800_53: '['SC.7']'
pci_dss: '['1.4']'
tsc: '['CC6.7', 'CC6.8']'

**Alert to be generated.
```

Activities

Wazuh Threat Hunting

Export Formatted 721 available fields Columns Density Fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 19, 2026 @ 09:53:43.509	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:47:16.466	windows	Software protection service scheduled successfully.	3	60642
Jul 19, 2026 @ 09:46:57.249	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:46:45.713	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:42:52.747	kali	Listened ports status (netstat) changed (new port opened or closed).	7	533
Jul 19, 2026 @ 09:42:36.549	windows	Windows Logon Success	3	60106
Jul 19, 2026 @ 09:41:59.312	furqan-virtual-machine	PAM: Login session closed.	3	5502
Jul 19, 2026 @ 09:41:59.312	furqan-virtual-machine	Wazuh server started.	3	502
Jul 19, 2026 @ 09:40:10.554	furqan-virtual-machine	PAM: Login session opened.	3	5501
Jul 19, 2026 @ 09:40:10.553	furqan-virtual-machine	Successful sudo to ROOT executed.	3	5402
Jul 19, 2026 @ 09:40:06.545	furqan-virtual-machine	PAM: Login session closed.	3	5502
Jul 19, 2026 @ 09:38:48.322	furqan-virtual-machine	Successful sudo to ROOT executed.	3	5402
Jul 19, 2026 @ 09:38:48.322	furqan-virtual-machine	PAM: Login session opened.	3	5501
Jul 19, 2026 @ 09:38:27.961	furqan-virtual-machine	pfSense DHCP Event Detected	5	100100
Jul 19, 2026 @ 09:38:27.959	furqan-virtual-machine	pfSense DHCP Event Detected	5	100100

Rows per page: 15 < 1 2 3 4 5 ... 197 >

3. Containment

After detecting the security events, the logs were reviewed to understand the nature of the activity. Since the detected events were related to DHCP requests and firewall block events generated during testing, no malicious activity was found. However, if suspicious traffic had been detected, pfSense firewall rules could be updated immediately to block the source IP address and prevent further communication.

Activities

furqan@furqan-virtual-machine: ~

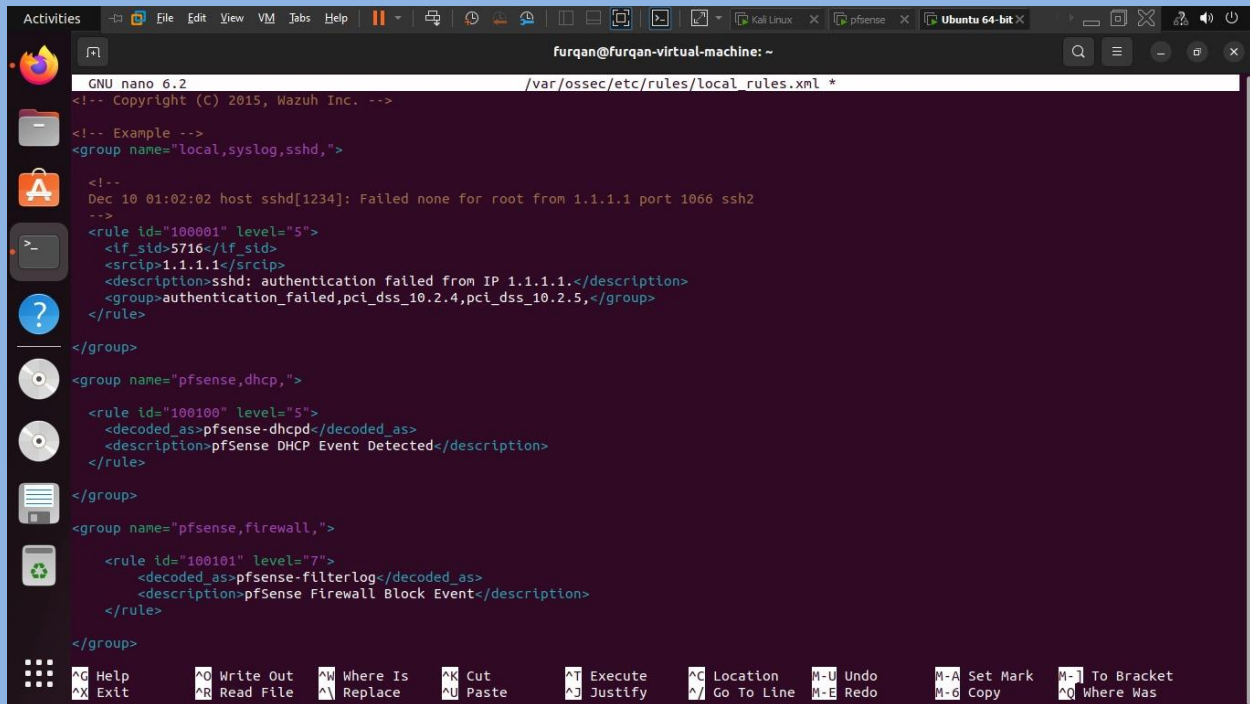
```

type one log per line
Jul 18 15:11:25 pfsense.home.arpa filterlog[33737]: 4,,1000000103,em1,match,block,in,4,0x0,,64,34744,0,DF,6,tcp,129,192.168.134.1,192.168.134.1,1514,50403,89,PA
**Phase 1: Completed pre-decoding.
full event: 'Jul 18 15:11:25 pfsense.home.arpa filterlog[33737]: 4,,1000000103,em1,match,block,in,4,0x0,,64,34744,0,DF,6,tcp,129,192.168.134.1,192.168.134.1,1514,50403,89,PA'
timestamp: 'Jul 18 15:11:25'
hostname: 'pfsense.home.arpa'
program_name: 'filterlog'
**Phase 2: Completed decoding.
name: 'pf'
action: 'block'
dstip: '192.168.134.1'
dstport: '50403'
id: '1000000103'
length: '89'
protocol: 'tcp'
srcip: '192.168.134.131'
srcport: '1514'
**Phase 3: Completed filtering (rules).
id: '87701'
level: '5'
description: 'pfSense firewall drop event.'
groups: '['pfsense', 'firewall_block']'
firedtimes: '1'
gpg13: '['4.12']'
hipaa: '['164.312.a.1']'
mail: 'False'
nist_800_53: '['SC.7']'
pcl_dss: '['1.4']'
tsc: '['CC6.7', 'CC6.8']'
**Alert to be generated.

```

4. Eradication

The custom decoder and rule were verified using the `wazuh-logtest` utility to ensure they were functioning correctly. Any configuration issues were resolved by updating the decoder and rule files and restarting the Wazuh Manager. After verification, alerts were successfully generated for the configured events.



The screenshot shows a terminal window titled "furqan@furqan-virtual-machine: ~" with the GNU nano 6.2 editor open. The file being edited is `/var/ossec/etc/rules/local_rules.xml`. The XML content defines three rule groups: "local,syslog,sshd", "pfsense,dhcp", and "pfsense,firewall". Each group contains one or more rules with specific IDs, levels, and descriptions. The first rule in the "local,syslog,sshd" group is triggered by a failed SSH login from 1.1.1.1. The second rule in the "pfsense,dhcp" group is triggered by a DHCP event. The third rule in the "pfsense,firewall" group is triggered by a firewall block event. The terminal window also shows a sidebar with various icons and a bottom status bar with keyboard shortcuts.

```
GNU nano 6.2 /var/ossec/etc/rules/local_rules.xml *
<!-- Copyright (C) 2015, Wazuh Inc. -->
<!-- Example -->
<group name="local,syslog,sshd">
  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>
</group>
<group name="pfsense,dhcp">
  <rule id="100100" level="5">
    <decoded_as>pfsense-dhcpd</decoded_as>
    <description>pfsense DHCP Event Detected</description>
  </rule>
</group>
<group name="pfsense,firewall">
  <rule id="100101" level="7">
    <decoded_as>pfsense-filterlog</decoded_as>
    <description>pfsense Firewall Block Event</description>
  </rule>
</group>
```

```
Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
<description>pfSense DHCP Event Detected</description>
</rule>
</group>
<group name="pfsense,firewall,">
  <rule id="100101" level="7">
    <decoded_as>pfsense-filterlog</decoded_as>
    <description>pfSense Firewall Block Event</description>
  </rule>
</group>
furqan@furqan-virtual-machine:~$ sudo /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.12.0
Type one log per line

Jul 18 01:29:22 pfsense.home.arpa dhcpd[91641]: DHCPDISCOVER from 00:50:56:c0:00:01 (DESKTOP-BUP9J5A) via em1

**Phase 1: Completed pre-decoding.
full event: 'Jul 18 01:29:22 pfsense.home.arpa dhcpd[91641]: DHCPDISCOVER from 00:50:56:c0:00:01 (DESKTOP-BUP9J5A) via em1'
timestamp: 'Jul 18 01:29:22'
hostname: 'pfsense.home.arpa'
program_name: 'dhcpd'

**Phase 2: Completed decoding.
name: 'pfsense-dhcpd'

**Phase 3: Completed filtering (rules).
id: '100100'
level: '5'
description: 'pfSense DHCP Event Detected'
groups: '['pfsense', 'dhcp']'
firedtimes: '1'
mail: 'False'

**Alert to be generated.
```

5. Recovery

After successful testing, all monitoring services were confirmed to be working correctly. The Wazuh Manager was running normally, firewall logs were continuously received, and alerts were successfully displayed in the Wazuh Dashboard. This confirmed that the monitoring environment had fully recovered and was ready for continuous operation.

Screenshot:

```
Activities  File Edit View VM Tabs Help  furqan@furqan-virtual-machine: ~
**Alert to be generated.
^C
furqan@furqan-virtual-machine:~$ sudo systemctl status wazuh-manager
wazuh-manager.service - Wazuh manager
Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
Active: active (running) since Sat 2026-07-18 15:30:57 PKT; 57min ago
Process: 28707 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
Tasks: 154 (limit: 4540)
Memory: 459.9M
CPU: 12min 20.985s
CGroup: /system.slice/wazuh-manager.service
└─28776 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
└─28777 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
└─28778 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
└─28781 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
└─28783 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
└─28826 /var/ossec/bin/wazuh-authd
└─28841 /var/ossec/bin/wazuh-db
└─28867 /var/ossec/bin/wazuh-execd
└─28880 /var/ossec/bin/wazuh-analysisd
└─28892 /var/ossec/bin/wazuh-syscheckd
└─28910 /var/ossec/bin/wazuh-remoted
└─28944 /var/ossec/bin/wazuh-logcollector
└─28964 /var/ossec/bin/wazuh-monitord
└─28987 /var/ossec/bin/wazuh-modulesd

15:30:46 18 جولي furqan-virtual-machine env[28707]: Started wazuh-analysisd...
15:30:47 18 جولي furqan-virtual-machine env[28707]: Started wazuh-syscheckd...
15:30:49 18 جولي furqan-virtual-machine env[28707]: Started wazuh-remoted...
15:30:51 18 جولي furqan-virtual-machine env[28707]: Started wazuh-logcollector...
15:30:52 18 جولي furqan-virtual-machine env[28707]: Started wazuh-monitord...
15:30:53 18 جولي furqan-virtual-machine env[28984]: 2026/07/18 15:30:53 wazuh-modulesd:router: INFO: Loaded router module.
15:30:53 18 جولي furqan-virtual-machine env[28984]: 2026/07/18 15:30:53 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
15:30:54 18 جولي furqan-virtual-machine env[28707]: Started wazuh-modulesd...
15:30:57 18 جولي furqan-virtual-machine env[28707]: Completed.
15:30:57 18 جولي furqan-virtual-machine systemd[1]: Started Wazuh manager.
furqan@furqan-virtual-machine:~$
```


6. Lessons Learned

This project provided practical experience in deploying a firewall, forwarding logs, integrating Wazuh SIEM, creating custom decoders and rules, and monitoring security events. It also improved understanding of centralized logging, real-time alert generation, and the incident response process. These skills are essential for performing the responsibilities of a SOC Analyst.

Incident Summary

Field	Details
Incident Type	Firewall and DHCP Event Monitoring
Detection Tool	Wazuh SIEM
Log Source	pfSense Firewall
Log Collection Method	Remote Syslog (UDP 514)
Custom Decoder	pfsense-dhcpd
Custom Rule	Rule ID 100100
Built-in Rule	Rule ID 87701
Status	Successfully Detected and Monitored